AP & RAP

Quiz 2

Date: 26 November 2025
Time allowed: 60 minutes **Full mark: 100**

1. (Definitions and examples, 20 points)
   (a) Provide the definition of a random variable; give two examples.
   (b) Provide the definition of independence for two σ-algebras and also independence for two random variables.
   (c) Provide the definition of a convex function.

2. (Short questions, 25 points) In the following questions, no justification is necessary. Simply provide the shortest possible correct response. Each question is worth 5 points.
   (a) Let X be a random variable and F be its distribution function. What are the properties of F that characterize it to be a distribution function?
   (b) Suppose F_1 and F_2 are both distribution functions of some random variables. Is $(F_1 + F_2)/2$ also distribution function of some random variable?
   (c) Let X be a random variable with $\mathbb{E}[|X|] < \infty$. Is it always true that $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$? Explain.
   (d) Suppose events A and B are independent. Are A^c and B^c also independent?
   (e) Suppose $A_1, A_2, \ldots$ is a sequence of events. Is it true that $\liminf_{n\to\infty} A_n \subset \limsup_{n\to\infty} A_n$?

3. (Theorems and proofs, 20 points) Please state Markov and Chebyshev inequalities and prove them.

4. (20 points) Let $(X_n : n \in \mathbb{N})$ be independent exponentially distributed random variables, i.e.,

$$P(X_n > x) = e^{-x}, \qquad x \geq 0.$$

   Then for $\alpha > 0$

$$P(X_n > \alpha \log n) = n^{-\alpha}.$$

   Let $A_n := \{X_n > \alpha \log n\}$, please prove:

$$P(\limsup_{n\to\infty} A_n) = \begin{cases} 0 & \text{if } \alpha > 1, \\ 1 & \text{if } \alpha \leq 1. \end{cases}$$

5. (15 points) Let X, Y be independent random variables taking values in $\mathbb{N}$ with

$$P(X = i) = P(Y = i) = \frac{1}{2^i}, \quad i = 1, 2, \ldots$$

Find the following probabilities:

(a) $P(\min(X, Y) \leq i)$, where $i \in \mathbb{N}$.

(b) $P(X = Y)$.

(c) $P(Y > X)$.